Phase 6: Project Testing

Model Evaluation & System Verification

Role	Name
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Model Evaluation

The XGBoost model was rigorously tested and optimized during the training phase usin
5-Fold Cross-Validation combined with RandomizedSearchCV.

- **Algorithm Used:** XGBoost Classifier
- **Optimization Technique:** RandomizedSearchCV (50 iterations, 5-Fold CV)
- **Best Cross-Validation Accuracy:** 0.8111 (81.11%)
- **Final Training Accuracy:** 1.0 (100%)
- **Final Testing Accuracy:** 0.7967 (79.67%)

Optimal Hyperparameters Identified:

Hyperparameter	Optimal Value
subsample	0.7
n_estimators	300
max depth	5
learning_rate	0.01
colsample bytree	1.0

Conclusion: The model achieved a perfect score on the training data and a strong 79.67% accuracy on unseen test data. The close proximity between the 5-Fold CV score (81.11%) and the Testing score (79.67%) proves that the model generalizes well and successfully avoids overfitting.

System Verification Cases

The deployed Flask application was tested with various inputs to ensure the data pipeli
(String → Int → DataFrame → Prediction) functions correctly.

Test Case	Input Sample	Expected Output	Actual Output	Status
Valid Input (Approved)	Gender=1, Income=5000, Credit_Hist=1	APPROVED	APPROVED	PASS
Valid Input (Rejected)	Gender=0, Income=1000, Credit_Hist=0	REJECTED	REJECTED	PASS
Boundary Test (Zero Income)	Income=0, CoappIncome=0	REJECTED	REJECTED	PASS
Invalid Input (Empty Field)	Income field left blank	Form Validation Error	Form Validation Error	PASS